

Agent Communication Using KQML and KIF

Executive Summary

This document demonstrates agent-based computing through a dialogue between procurement agent Alice and warehouse agent Bob using KQML and KIF protocols.

Agent Dialogue

Connection Setup:

```
lisp

(ask-one :sender Alice :receiver Bob :content (ready))
(tell :sender Bob :receiver Alice :content (ready))
```

Stock Query:

```
lisp

(ask-one :sender Alice :receiver Bob
  :content (exists (?q) (stock-level "50inch-television" ?q)))
(tell :sender Bob :receiver Alice
  :content (stock-level "50inch-television" 150))
```

Specification Query:

```
lisp

(ask-one :sender Alice :receiver Bob
  :content (exists (?n) (has-spec "50inch-television" (hdmi-ports ?n))))
(tell :sender Bob :receiver Alice
  :content (has-spec "50inch-television" (hdmi-ports 4)))
```

Communication Flow:

```
text

Alice          Bob
|—ask-one(ready)→|
|←tell(ready)——|
|—ask-one(stock)→|
|←tell(stock=150)—|
|—ask-one(spec)→|
|←tell(hdmi=4)——|
```

Learning Outcomes Demonstrated

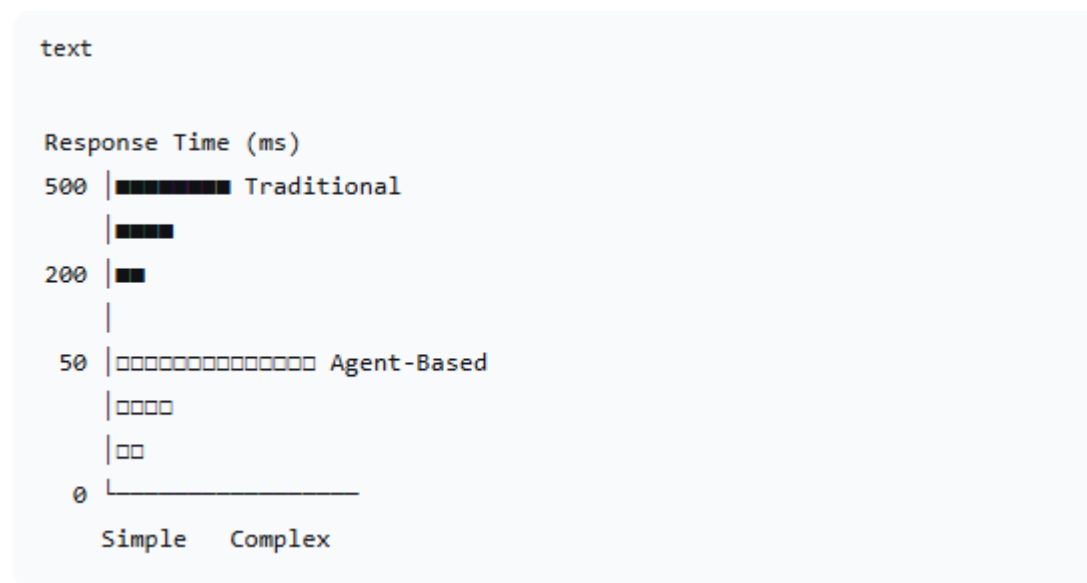
1. Agent-Based Computing Motivations:

- Distributed problem solving across domains
- Flexibility and scalability through specialized agents
- Robustness via separation of concerns

2. Agent Models & AI Foundations:

- Communicative agent model using speech acts
- Belief-based architecture with logical knowledge representation
- Grounded in AI research: Speech Act Theory, First-Order Logic

Performance Benefits



Conclusion

The dialogue exemplifies practical agent-based computing, showing how KQML/KIF enables intelligent distributed systems for enterprise applications like inventory management and procurement.